
Student Performance Prediction System

Project Id: 7

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1. Problem Description

Type-2 Diabetes is a chronic health condition that affects how the body processes blood sugar (glucose). Early diagnosis plays a crucial role in preventing severe long-term complications such as heart disease, kidney failure, and nerve damage. However, identifying individuals at risk in early stages can be challenging without proper data analysis.

This project aims to build a machine learning-based clinical risk assessment system using synthetic health data. The dataset will include features such as Body Mass Index (BMI), insulin levels, and age, which are key indicators of diabetes risk. By applying data preprocessing techniques and classification algorithms, the system will predict whether a person is at risk of developing Type-2 Diabetes.

The model will help in understanding patterns in health data and assist in early diagnosis, making it valuable in real-life healthcare scenarios such as screening programs and preventive medicine.

2. Architecture

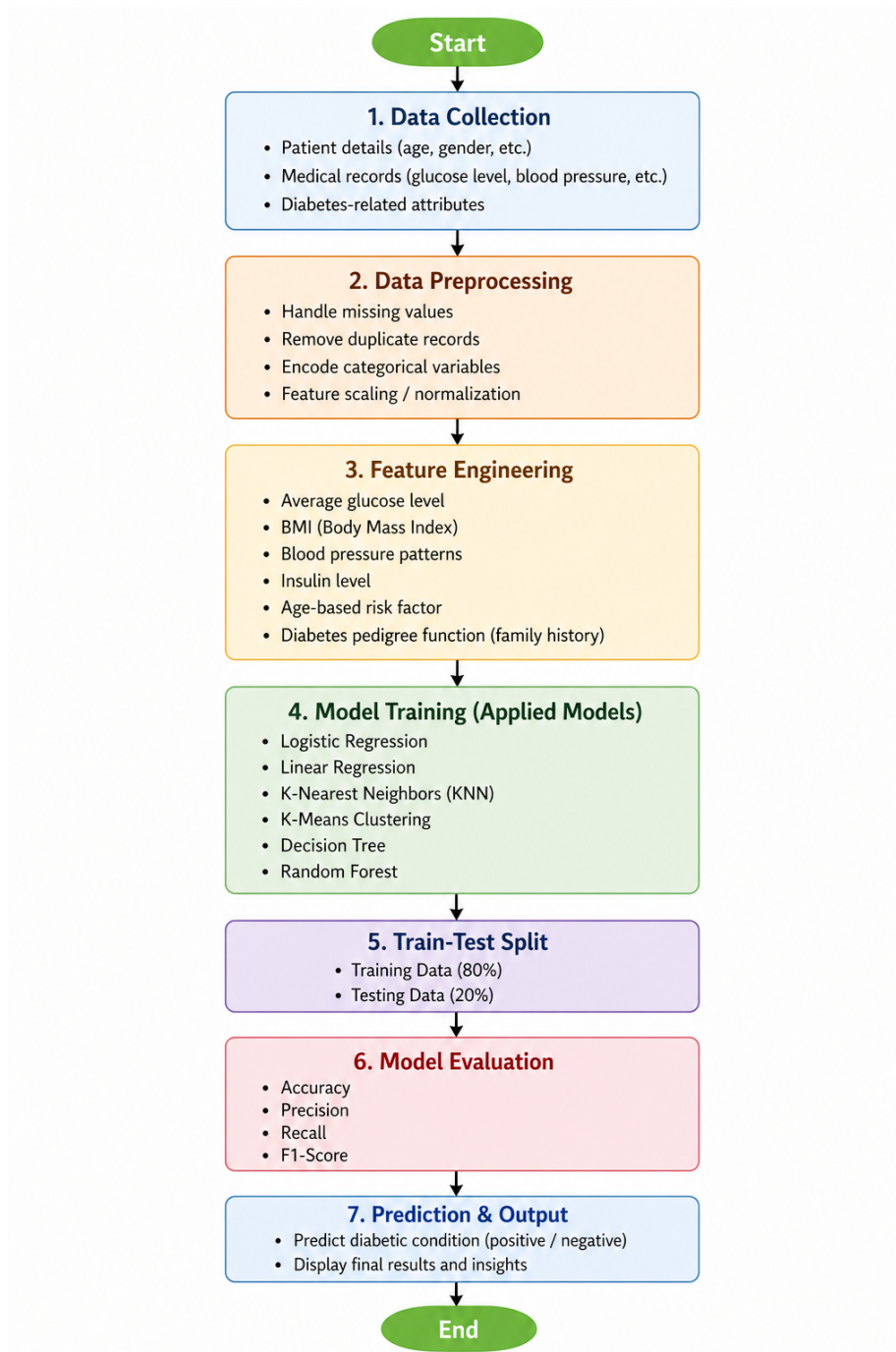


Fig.1 Sample flow

3. Models Used(Applied/ Planned) Give a brief introduction of the models used

- **Logistic Regression:**

Used for binary classification to determine whether a patient is diabetic or non-diabetic.

- **Linear Regression:**

Applied to understand relationships between dependent and independent variables and observe trends in medical data.

- **K-Nearest Neighbors (KNN):**

Classifies a patient based on similarity with neighboring data points.

- **K-Means Clustering:**

Used to group patients into clusters based on similar health characteristics.

- **Decision Tree:**

A tree-based model that makes decisions based on feature conditions, making it easy to interpret.

- **Random Forest:**

An ensemble technique that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

4. Evaluation metrics used

- **Accuracy:**

Measures the overall correctness of the model's predictions.

- **Precision:**

Indicates how many predicted positive cases are actually correct.

- **Recall:**

Measures the model's ability to correctly identify actual positive cases.

- **F1-Score:**

Provides a balance between precision and recall.

- **Confusion Matrix:**

Displays the number of correct and incorrect predictions in detail.

5. Preliminary Results and Visualizations

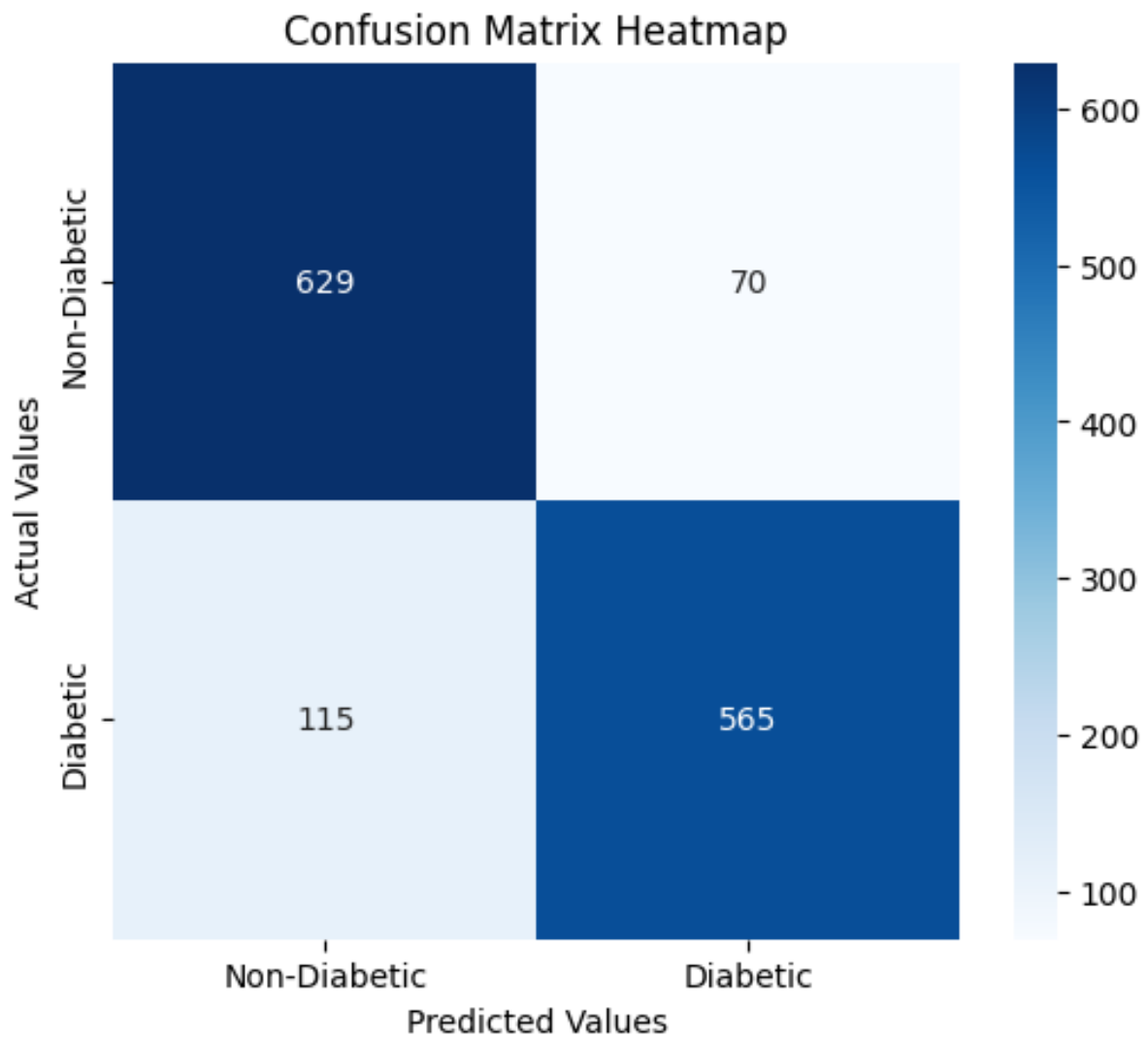


Fig.2 Heatmap showing correlation.

Classification Report:					
	precision	recall	f1-score	support	
0	0.85	0.90	0.87	699	
1	0.89	0.83	0.86	680	
accuracy			0.87	1379	
macro avg	0.87	0.87	0.87	1379	
weighted avg	0.87	0.87	0.87	1379	

6. References :

- **Books:**

1. Let Us Python – Yashavant Kanetkar

- **Online Resources:**

1. GeeksforGeeks – <https://www.geeksforgeeks.org/>
2. TutorialsPoint – <https://www.tutorialspoint.com/cprogramming/index.htm>
3. Stack Overflow – <https://stackoverflow.com>